

AUSTIN AKERLEY

SENIOR SOFTWARE ENGINEER



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WORK EXPERIENCE

Microsoft - PKI Team - Senior SWE

- **Product:** My team is responsible for critical tier-4 datacenters that host Microsoft's Public Key Infrastructure (MSPKI), CodeSign, and Malware Scanning services.
- **Primary Role:** Design and scale the PKI infrastructure of our core security services while improving cryptographic security of our product by utilizing TPMs.
- **Secondary Role:** Datacenter buildout, security, and maintenance. Managing over 1000 physical servers across 3 datacenters for the highest security customers.

Oracle Cloud(OCI) - KMS - SWE II

- **Product:** Key Management Service (KMS) is a cloud service that manages the encryption keys used to protect sensitive data.
- **Primary Role:** Hardware Security Module(HSM) integration expert specializing in **C/C++** to integrate vendor-provided HSM cards as the core foundation for trust and security in KMS. Led the orchestration of HSM changes across 500 bare-metal hosts.
- **Secondary Role:** Implemented cryptographic algorithms for KMS APIs in **Java** for an upcoming small-footprint dedicated cloud offering.
- **Smaller Projects:**
 - Built developer tool in **Python** for validating OCI alarm infra, it is now a required step in new OCI cloud service creation.

Microsoft O365 - Teams - SRE II

- **Product:** Dedicated Top Secret gov. **Azure** cloud region.
- **Primary Role:** Top Secret SCI w/ Poly cleared SRE. Deployed the infrastructure and application for MS Teams to 2 dedicated cloud regions using **IaC** & **automation**.
- **Secondary Role:** Deployed the infrastructure and application for Azure DevOps pipelines in the secure cloud regions. Automated tasks with **Powershell** and **Python**.
- **Smaller Projects:**
 - Debugged app and infra issues for 73 micro-services across 15 teams and implemented region-specific fixes.
 - Contributed heavily to region-specific operational **documentation** and onboarding documentation.

Lockheed Martin Space - Cyber SWE

- **Product:** A custom Network Interface Device(NID) with a Xilinx System on Chip(**SoC**) **FPGA** for the F35 fighter jet.
- **Primary Role:** **System Verilog** developer implementing our primary FPGA cyber components.
- **Secondary Role:** Completed major feature development for our SoC **kernel modules** in **C++**.

EDUCATION

University of Washington

Masters of Science*
Computer Science



*Currently enrolled - graduating April 2027

Texas A&M University

Bachelors of Science
Computer Engineering



Minors

Cyber Security
Mathematics

Paid Research Assistant

- Created agricultural software for Noble Research Institute.
- **Enjoy pecans?** Ask me how I may have prevented you from eating a bad one.

SKILLS

- **Cryptography** and **security** specialist
- Experience with creating, maintaining, and monitoring highly **distributed systems** that require **high availability**
- Proficient in **state management** design
- **Cloud expert** with experience designing cloud native applications and automation
- Experience with debugging and tools such as **valgrind** & **gdb**
- Experienced with customer facing front end technologies such as **Expo**, **Tailwind**, **Node JS**, **React Native**, and **Typescript**
- Experienced with modern back end technologies such as **Python**, **Django**, **T3**, **GraphQL**, **JWT OAuth**, **AWS**, **OCI**, and **Azure**

LANGUAGES

Python	C++ & C	C#
Java	Linux Shell	HDLs
Typescript	IaC/Terraform	HTML/CSS